

Nicholas Ketz, PhD.

Applied Research Scientist: Artificial Intelligence, Bioinformatics, Mathematical Models, Human Studies
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I am an applied scientist developing machine learning solutions to advanced research problems. I have more than 10 years of experience in both academic and industrial research labs. I have experience individually contributing, leading small teams and coordinating across multiple departments throughout the entire AI product life cycle.

Relevant Experience

- Large Language Modeling** 03/2022 - Present
- DNA Large Language Modeling applied to cell and gene therapy optimization and cross-species engineering
- Generative AI** 01/2020 - 01/2022
- Created AI systems for generating video and images grounded in learned human concepts and semantic understanding
- Model Based Reinforcement Learning** 01/2020 - 01/2022
- Developed Agent-based domain adaptation techniques within autonomous driving using model-based reinforcement learning
- Lifelong Deep Learning** 09/2018 - 01/2021
- DARPA L2M key personnel, researched and implemented methods for sequential multitask learning in deep neural networks
- Closed-loop Neural Stimulation** 09/2016 - 06/2018
- Developed devices and algorithms for closed-loop neural stimulation in human subjects to enhance cognitive performance

Education/Employment

- Principal AI/ML Scientist** *Colossal Biosciences/Form Bio* 02/2022 - Present
- Advise and review team in the full AI life cycle using genomic, conservation and biological data
- Research Scientist: Information and Systems Sciences** *HRL Laboratories* 09/2016 - 01/2022
- Basic AI research funded by Department of Defense and product development for aerospace and automotive industries
- PhD: Computational Cognitive Neuroscience** *University of Colorado, Boulder* 09/2010 - 09/2016
- Development of biologically inspired neural networks informed by neuro-imaging experiments of cognitive function
- Research Assistant: Davachi Human Memory Lab** *New York University* 09/2007 - 07/2010
- Supported graduate level research for neuro-imaging of human long-term memory system
- BA: Physics, minor: Psychology** *University of Minnesota, Twin Cities* 09/2003 - 06/2007
- Studied physics with a minor in psychology, with research experience in particle physics and cognitive-neuroscience

Skills

- **Technical:** Deep Learning (CNN, RNN, Transformers, RL), Machine Learning (Optimization, Unsupervised Learning), Statistics (Bayesian, Parametric/Non-parametric, Time-series), Programming (PyTorch, TensorFlow, scikit-learn, Docker, Git, Bash, SQL), Agentic AI (LLM agents, tool-use orchestration, agentic coding workflows), Cloud (GCP)
- **AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, agentic development workflows for code and product delivery, containerized system design, CI/CD pipelines
- **Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents/White Papers

- System and method for semantically grounded video generation** *US Patent, 2024*
- Generative video AI system prompted by learned human concepts
- A domain-agnostic approach for characterization of lifelong learning systems** *Neural Networks, 2023*
- DoD project for evaluating neural network systems across multiple domains and tasks
- Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures** *White Paper, 2022*
- Extracting learned motifs from DNA sequence models predictive of gene therapy manufacturing failures
- Sliced Cramer synaptic consolidation for preserving deeply learned representations** *Spotlight Paper ICLR, 2020*
- Framework for overcoming catastrophic forgetting by preserving the distribution of the network's output at an arbitrary layer